

Official Agricultural Extension Advisory: Rice Blast in Rice (Paddy)

Causal Pathogen: Magnaporthe oryzae (anamorph Pyricularia oryzae)

Issuing Source: ICAR - National Rice Research Institute (NRRI) Disease Compendium | Date: 2024-03-15

1. Identification and Morphological Symptoms

Rice blast affects all aerial parts of the rice plant. On leaves, symptoms appear as spindle-shaped or diamond-shaped spots with gray or white centers and distinct brown or reddish-brown margins. Under humid conditions, older lesions enlarge, coalesce, and cause complete leaf desiccation (leaf blast). At later stages, collar rot appears at the leaf junction, and neck blast attacks the neck node of the panicle, turning it dark brown to black and leading to chaffy grains and lodging.

2. Environmental Epidemiology & Causes

Favored by prolonged periods of leaf wetness (>10 hours), high relative humidity (>90%), moderate night temperatures (18-24 deg C), overcast skies with light rain or drizzle, and excessive application of nitrogenous fertilizers without adequate potassium balance.

3. Cultural Management and Field Prevention

Prevention: Avoid excessive and staggered nitrogen fertilizer application; split nitrogen into three or four doses. Maintain optimal plant density and ensure good soil drainage. Treat seed prior to sowing to eliminate seed-borne inoculum.

Cultural Practices: Destroy crop residues and volunteer rice plants after harvest. Practice balanced fertilization with NPK ratio of 4:2:1 or as per soil test recommendations, with split nitrogen application. Maintain field sanitation and remove susceptible grass weed hosts from bunds.

4. Organic and Biological Management

Seed treatment with *Pseudomonas fluorescens* @ 10 g/kg of seed, followed by nursery root dip in 2.5 kg/ha *P. fluorescens* suspension. Foliar spray of *Pseudomonas fluorescens* formulation @ 2.5 kg/ha or *Trichoderma harzianum* @ 5 g/L at early tillering and pre-flowering stages. Application of neem cake to soil @ 150 kg/ha.

5. Verified Chemical Interventions

Seed treatment with Tricyclazole 75% WP @ 2.0 g/kg seed. In the main field, foliar spray of Tricyclazole 75% WP @ 0.6 g/L or Azoxystrobin 18.2% + Difenconazole 11.4% SC @ 1.0 mL/L or Isoprothiolane 40% EC @ 1.5 mL/L at the first appearance of leaf blast lesions, repeating at booting/heading if neck blast weather persists.

• **Active Ingredient:** Tricyclazole 75% WP | **Reference Formulation:** Beam (registered standard formulation) | **Registrant:** Corteva Agriscience (registered registrant) | **Source:** DPPQS CIBRC Registered Fungicides List (Govt of India) (2024-01-10)

• **Active Ingredient:** Isoprothiolane 40% EC | **Reference Formulation:** Fuji-one (standard reference) | **Registrant:** Nihon Nohyaku Co. (registered registrant) | **Source:** ICAR-NRRI Technical Advisory 2024 (2024-02-18)

Notice: Chemical recommendations are derived strictly from central agricultural research advisories. Farmers must consult local district agricultural extension officers for local dosage and label claims.